Course Code : ECT 354 **ISJT/MW – 23 / 1808**

Fifth Semester B. Tech. (Electronics and Communication Engineering) Examination

DIGITAL SIGNAL PROCESSING

Time : 3 Hours] [Max. Marks : 60

Instructions to Candidates :—

(1) All questions are compulsory.

(2) Use of Non – programmable calculator is allowed.

(3) Assume suitable data wherever necessary.

1. (a) Compute the circular convolution of two sequences using DFT-IDFT method.

$\mathbf{x(n) = \{1, 1, 2, 1\}}$ and $\mathbf{h(n) = \{1, 2, 3, 4\}}$ 5(CO1,2)

(b) Compute 8-Point DFT of sequence $\mathbf{x(n) = \{1, 1, 1, 1, 1, 1\}}$. Use Radix-2, DIT-FFT Algorithm. 5(CO1,2)

2. (a) Obtain the direct form I, direct form II, cascade and parallel form realization for the system :

$$y(n) = -0{\cdot}1\, y(n-1) + 0{\cdot}2\, y(n-2) + 3x(n) + 3{\cdot}6\, x(n-1) + 0{\cdot}6\, x(n-2)$$

8(CO1,3)

(b) Realize the given system function using minimum number of multipliers :

$$H(z) = 1 + 0{\cdot}25\,z^{-1} + 0{\cdot}75\,z^{-2} + 0{\cdot}75\,z^{-3} + 0{\cdot}25\,z^{-4} + z^{-5}$$ 2(CO1,3)

3. Design and realize digital IIR Butterworth LPP filter using bilinear transformation with following constraints. Assume T = 1 sec.

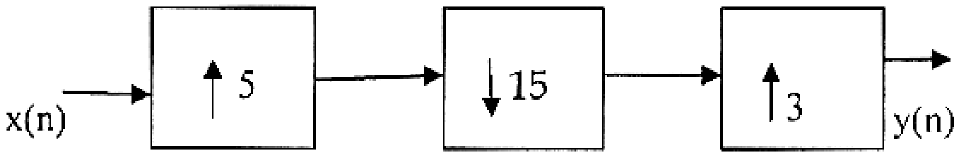
$$0{\cdot}707 \leqslant |H(e^{jw})| \leqslant 1 \quad \text{for} \quad 0 \leqslant w \leqslant \pi/2$$

$$|H(e^{jw})| \leqslant 0{\cdot}2 \quad \text{for} \quad 3\pi/4 \leqslant w \leqslant \pi$$ 10(CO1,3)

4. Design a high pass filter using Hamming window with a cut-off frequency of 1·2 rad/sec and $N = 9$.
Consider a Hamming Window given by

$$W_H(n) = 0{\cdot}54 - 0{\cdot}46 \cos(2\pi\, n/(N-1)), \quad 0 \leq n \leq N-1$$ 10(CO1,3)

5. (a) Obtain the expression for the output $y(n)$ in terms of input $x(n)$ for the multirate system given as follows :



5(CO1,4)

(b) State various applications of multirate signal processing. Explain any one application in detail. 5(CO1,4)

6. (a) What are the on chip peripherals available on programmable digital signal processors ? Explain their functions with block diagram. 5(CO1,5)

(b) What are the advantages and applications of DSP processors over conventional microprocessors ? 5(CO1,5)